

MyIntern: A Centralized Role-Based Web Platform for Internship Coordination and Job Matching in Higher Education

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Abstract—This study presents the development of *MyIntern*, a centralized web-based platform tailored for managing On-the-Job Training (OJT) processes in the College of Computing and Information Sciences at Caraga State University. Addressing inefficiencies in the manual OJT process, *MyIntern* integrates features such as a resume builder, job matching algorithm, electronic document submission, progress tracking, and role-specific dashboards for students, coordinators, and employers. Developed using the Laravel PHP framework, the system underwent usability testing involving students, employers, and a coordinator, with results showing good usability (76–83.21%). The platform demonstrated significant improvements in communication, submission compliance, and overall OJT coordination, confirming its effectiveness and scalability for institutional use.

Keywords— OJT management, internship system, job matching, web-based platform, usability evaluation, Laravel, academic coordination

I. INTRODUCTION

As universities embrace digital transformation, more and more academic institutions are turning to web-based platforms to simplify both administrative and academic tasks. One area that has lagged behind, however, is the management of On-the-Job Training (OJT) programs especially within computing-related disciplines. These internships play a critical role in bridging classroom knowledge with real-world experience, yet many institutions in the Philippines, including the College of Computing and Information Sciences (CCIS) of Caraga State University, still rely on outdated, manual processes.

Currently, students often submit application forms and requirements through printed documents or scattered messages in group chats. Coordinators track progress through spreadsheets or paper records, and announcements are made via informal channels. This system is not only time-consuming

but also prone to miscommunication, misplaced files, and tracking errors. For coordinators managing hundreds of students, it becomes overwhelming. Students, on the other hand, face uncertainty in finding internship placements that suit their skills and sometimes submit documents late or only after completing their OJT.

These challenges highlight the urgent need for a centralized and automated system. Recent studies support the benefits of digitizing internship processes, citing improvements in communication, record management, and overall efficiency [1], [2]. Systems that incorporate features such as resume builders [3], job matching algorithms [4], and digital document handling [5] have shown measurable impact on stakeholder satisfaction and process usability. Furthermore, principles from Human-Computer Interaction (HCI) [6] and job matching theory [7] provide a solid foundation for designing systems that are both intuitive and effective.

This paper introduces *MyIntern*, a centralized, web-based OJT management platform developed to address these specific gaps. Built using the Laravel framework, the system offers role-based access and functionality tailored to students, coordinators, and company supervisors. Through features like automated job matching, real-time progress tracking, digital requirement submission, and analytics dashboards, *MyIntern* aims to streamline the OJT experience and provide a scalable solution that academic institutions can adopt and expand.

II. RELATED LITERATURE

Internship management systems have become increasingly relevant as academic institutions look to strengthen industry engagement and prepare students for the workforce. In the Philippine context, however, OJT coordination still heavily relies on manual approaches making it difficult to manage records efficiently or maintain consistent communication [8].

[9] Hingpit et al. (2024), observed that many universities continue to use printed forms, email, and group chats to manage OJT-related processes. This fragmented setup creates

administrative bottlenecks and frustrates both students and coordinators. [10] Casugay et al. (2024), highlighted how digitizing academic documents improves transparency, ensures record accuracy, and simplifies compliance tracking.

Several studies have underscored the importance of integrated resume-building tools. [11] Tyagi et al. (2020), found that students who used guided templates were more confident and better prepared for internship applications. Similarly, [12] Kanjalkar et al. (2024), introduced the use of intelligent student profiling and job matching systems that align skills and academic history with company requirements leading to more relevant and successful placements.

Automation also plays a key role in improving system efficiency. [13] Perri-Moore et al. (2016), noted that real-time alerts and validation checks reduced coordinator workload and improved accountability. [14] Finata et al. (2019), demonstrated that digital dashboards allowed instructors to monitor student progress more effectively and intervene when necessary.

Several commercial platforms like JobStreet and Kalibrr offer resume management and job posting capabilities, but they are not tailored for academic OJT workflows. They lack features such as document compliance tracking, internship evaluations, and role-based access. Practicum360 offers robust monitoring tools but comes at a high cost, making it less suitable for public institutions [15]. Résumatcher provides basic compliance tracking but lacks integrated job matching and a centralized communication framework [16].

Collectively, these findings underscore the necessity of developing *MyIntern* a system that merges best practices from these studies while addressing the contextual needs of Philippine state universities through localization, affordability, and integration of role-based features.

III. METHODOLOGY

The design of *MyIntern* follows a structured and modular approach to address the inefficiencies in the manual handling of On-the-Job Training (OJT) at the College of Computing and Information Sciences (CCIS), Caraga State University. The platform is built to support role-specific functionalities for students, coordinators, and company supervisors streamlining the entire internship lifecycle from application to evaluation.

A. Conceptual Framework

Fig. 1. illustrates how the system digitizes the OJT process, starting from public job listings on the landing page and progressing through user registration, profile setup, resume building, job matching, requirement submission, and progress tracking. The centralized platform processes all user inputs and interactions to generate real-time updates, compliance analytics, and internship monitoring across all roles.

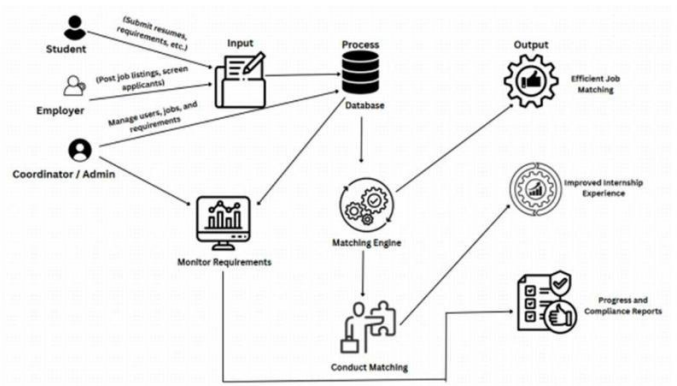
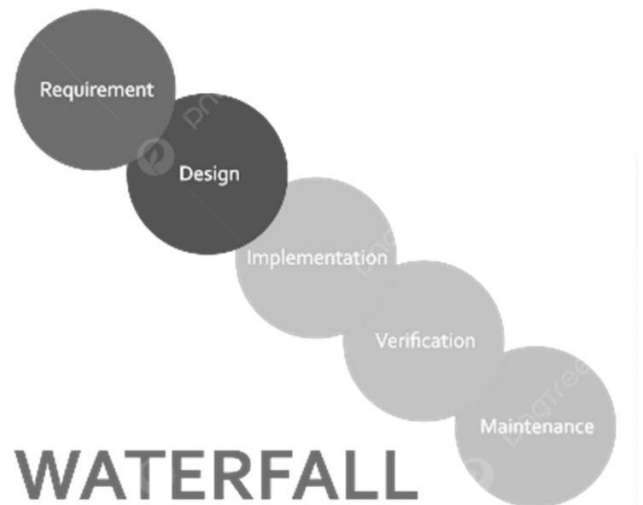


Fig. 1. Conceptual Framework of the Study

B. System Development Life Cycle



To ensure a structured development process, the study adopted the Waterfall Model of the Systems Development Life Cycle (SDLC). This methodology ensures thorough planning and documentation from the outset, promoting efficiency and accountability. It allows the research team to clearly define deliverables at each phase and ensures that each milestone is completed before moving on to the next.

C. System Architecture and Roles

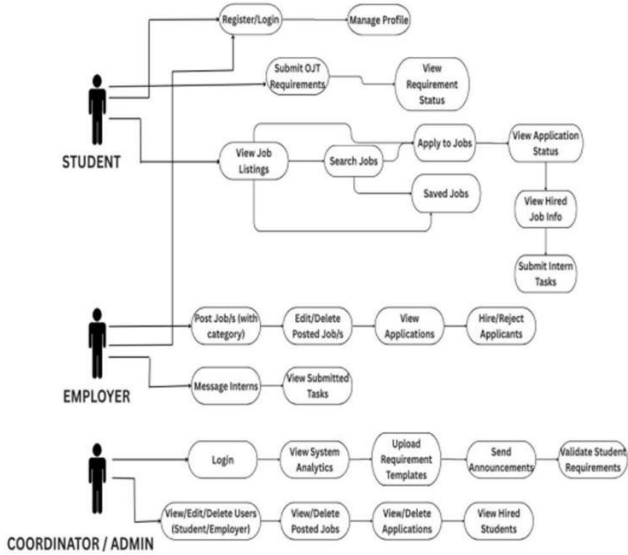


Fig. 2. Use case diagram of the study

The system was designed to accommodate three main user roles, each with dedicated interfaces and functionalities:

- **Students:** Can register, build resumes, search and apply for jobs, submit requirements, and track their internship progress.
- **Coordinators (Admins):** Oversee student records, validate submissions, monitor OJT status, manage announcements, and generate reports.
- **Company Representatives:** Post job listings, review applications, monitor intern activities, and provide evaluations.

D. Functional and Non-Functional Requirements

Functional Requirements include:

- Public job listing (view-only without login)
- Role-based login and dashboard access
- Resume builder with dynamic forms
- Skill-based job matching engine
- Document upload and approval workflow
- Internship task monitoring
- Admin analytics dashboard

Non-Functional Requirements include:

- **Usability:** Simple, responsive UI using Bootstrap
- **Security:** Encrypted credentials and role-based access
- **Performance:** Optimized queries and fast load times
- **Scalability:** Modular design for future expansion
- **Maintainability:** Follows Laravel MVC pattern for code clarity

E. Hardware and Software Requirements

The platform was developed using:

Backend:	PHP (Laravel Framework)
Frontend:	Blade Template Engine, Bootstrap, JavaScript
Database:	PostgreSQL
Server:	Apache HTTP Server 2.4
OS Compatibility:	Windows 10 or macOS Big Sur
Processor	Intel Core i5 minimum
RAM	8GB DDR4
Storage	500GB SSD

F. Testing and Deployment

Upon completion of the development phase, the system underwent a two-stage testing process: Alpha Testing and Usability Testing.

• Alpha Testing

An internal round of alpha testing was conducted by selected students, coordinators, and simulated employer accounts. The purpose was to test end-to-end flows like creating resumes, posting jobs, applying, and transitioning hired students to task monitoring. Bugs and inconsistencies were logged and resolved.

• Usability Testing

TABLE I. USABILITY TESTING QUESTIONNAIRE FOR STUDENTS

Criteria	Statement	SA-4	A-3	D-2	SDA-1
Interface & Navigation	The student dashboard is easy to navigate.				
	The process of setting up my profile is straightforward.				
Resume Builder & Job Matching	The resume builder is user-friendly and helpful.				
	The job matching feature displays relevant job postings.				
Application Process	I was able to apply to jobs without confusion.				
	The status of my applications was clear and easy to track.				
Requirement Submission	Uploading OJT requirements was simple.				
	I received feedback or updates on my submitted requirements.				
Overall Experience	The system made my OJT process easier.				
	I am satisfied with the overall functionality for students.				

TABLE II. USABILITY TESTING QUESTIONNAIRE FOR COMPANY/EMPLOYERS

Criteria	Statement	SA-4	A-3	D-2	SDA-1
Interface & Profile Setup	The employer dashboard is easy to navigate.				
	Creating and updating my company profile was simple.				
Job Posting & Applications	Posting a job with category selection is straightforward.				
	I was able to easily view student applications.				
Intern Monitoring	Hiring or rejecting applicants was efficient.				
	I could view tasks submitted by interns clearly.				
Overall Experience	Messaging interns through the system was effective.				
	The system helps streamline the internship recruitment process.				
	I am satisfied with the system features provided for employers.				

TABLE III. USABILITY TESTING QUESTIONNAIRE FOR ADMIN/COORDINATOR

Criteria	Statement	SA-4	A-3	D-2	SDA-1
Admin Dashboard & Access	The admin dashboard is accessible and well-structured.				
	I was able to manage student and employer accounts with ease.				
Content & Requirement Management	Uploading requirement templates was smooth.				
	Sending announcements was clear and effective.				
Monitoring & Analytics	The system provides helpful insights into student's activity through the system.				
	Viewing hired students and their status is simple.				
System Control	I had sufficient control over jobs, applications, and system data.				
Overall Experience	The system enhances OJT management for coordinators.				
	I am satisfied with the tools available to administrators.				

The images shown here illustrates a sample questionnaire that will be used in this research after the application has been subjected to alpha testing. The questionnaire adopts the Likert scale method to measure the different factors of usability. The Likert scale utilizes four levels, which include Strongly Agree (SA), Agree (A), Disagree (D) and Strongly Disagree (SDA). To compute the outcomes, the researchers can make use of the formula for averages that is regularly utilized in other types of usability tests. To determine the average, follow the formula provided below:

$$\text{Average} = \frac{\text{Sum of all responses}}{\text{Total number of responses}} \times 100$$

After obtaining all the results, the researchers will analyze whether the system is perceived as easy or difficult to use. The criteria presented below outline the descriptive analysis of the obtained results, including an assessment of the effectiveness of our system.

TABLE IV. USABILITY LEVELS

Numerical Scale	Descriptive Rating	Quantitative Description
4	Strongly Agree (SA)	Exhibits 91-100% usability of the method
3	Agree (A)	Exhibits 61-90% usability of the method
2	Disagree (D)	Exhibits 31-60% usability of the method
1	Strongly Disagree (SDA)	Exhibits 1-30% usability of the method

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G. Role-Based Access and User Management

The system implements role-based dashboards for three types of users—students, coordinators, and company representatives. Registration forms include a role-selection mechanism, which controls access to specific modules and features. Coordinators (admins) are manually registered by developers to prevent unauthorized administrative access.

Each user group interacts with the system through custom-tailored dashboards:

- **Students** access tools for resume creation, job search, requirement submission, and internship progress tracking.
- **Employers** manage job listings, review applicants, and monitor hired interns' activities.
- **Coordinators** can validate documents, view analytics, upload templates, and send announcements.

H. Resume Builder Module

The system features a dynamic, step-by-step resume builder. It collects personal data, academic background, skills, and category preferences, which are critical for job matching. Laravel's `old()` helper retains form inputs during validation, and predefined dropdowns ensure data consistency. This structured input directly feeds into the job matching algorithm.

I. Job Matching Algorithm

The matching logic uses category-based prioritization. Student resume categories are compared with job listing tags using Laravel's `orderByRaw()` method to display matching jobs first. This ensures students see relevant internships without excluding additional opportunities, striking a balance between precision and flexibility.

J. Requirement Submission and Validation

The platform provides a clear and secure submission interface for OJT-related documents. Students upload files directly under posted requirements, with the system enforcing format validation, deadline tracking, and Cross-Site Request Forgery (CSRF) protection. Each file is mapped to a specific requirement ID for accurate tracking and verification.

K. Employer progress monitoring

Employers can track weekly intern activities through a progress monitoring dashboard. Submitted tasks are grouped by week using the Carbon library and are paginated to maintain performance. This allows supervisors to quickly assess student performance and provide evaluations based on structured outputs.

L. Admin dashboard and Analytics

The administrator dashboard aggregates key metrics such as the number of registered users, employers, job posts, applications, and hired students. Visualizations include a monthly line chart (for student hiring trends) and a doughnut chart (for system usage distribution), powered by Chart.js. These tools assist coordinators in strategic decision-making.

M. Document Verification and Monitoring

Admins have access to a live monitoring page for validating uploaded student documents. Each submission includes user identity, file metadata, and timestamps. The system enables direct downloads and secure access for authorized reviewers, improving audit readiness and workflow compliance.

IV. RESULTS AND DISCUSSION

To assess the usability, effectiveness, and satisfaction of the *MyIntern* system, a structured evaluation was conducted through alpha testing and formal user testing sessions involving key stakeholders from the College of Computing and Information Sciences (CCIS), Caraga State University. The evaluation adopted both quantitative and qualitative methods to validate system performance.

A. Testing Procedure

Testing was conducted in two phases:

1. **Alpha Testing** – Internal testing was performed by selected users simulating full workflows (registration, job posting, application, document submission, and task tracking). Issues such as validation errors and user flow inconsistencies were logged and resolved.
2. **Formal Usability Testing** – On March 17, 2025, sixteen participants (10 students, 5 company/department supervisors, and 1 OJT coordinator) evaluated the system through real-world usage scenarios. Tasks included registration, job matching, uploading requirements, and tracking progress.

A post-test questionnaire was distributed using a **four-point Likert scale** measuring three dimensions:

- **Effectiveness**
- **Efficiency**
- **Satisfaction**

Scores were computed as percentages and interpreted using predefined criteria:

- 91–100%: Highly favorable usability
- 61–90%: Good usability
- Below 61%: Needs improvement

B. Student Evaluation Results

Ten student participants interacted with the platform's core modules including resume building, job application, and requirement submission. The average usability scores were:

TABLE V. STUDENT EVALUATION RESULT

Criterion	Total Score	Max Score	Percentage
Effectiveness	230	280	82.14%
Efficiency	229	280	81.79%
Satisfaction	233	280	83.21%

Students appreciated the structured flow and real-time feedback, though some noted interface areas for improvement.

C. Company/Department Supervisor Evaluation

Five company or department representatives used the employer dashboard to post jobs, monitor applicants, and evaluate intern submissions. The results were:

TABLE VI. SUPERVISOR EVALUATION RESULT

Criterion	Total Score	Max Score	Percentage
Effectiveness	82	100	82%
Efficiency	78	100	78%
Satisfaction	82	100	82%

Supervisors found the platform practical but recommended more advanced filtering and task tracking tools.

D. Coordinator Evaluation

The OJT coordinator assessed the admin dashboard, submission monitor, and analytics modules.

TABLE VII. COORDINATOR EVALUATION RESULT

Criterion	Total Score	Max Score	Percentage
Effectiveness	79	100	79%
Efficiency	76	100	76%
Satisfaction	78	100	78%

The coordinator valued centralized access and simplified document tracking, suggesting additional tools like bulk actions and real-time messaging.

E. Summary and Interpretation

Overall, *MyIntern* achieved average usability scores ranging from **76% to 83.21%** across all roles, falling within the "Good usability" range. Users highlighted significant improvements in submission compliance, tracking, and reduced communication delays compared to the previous manual system.

While the platform was deemed ready for deployment, feedback also indicated future enhancements such as mobile responsiveness, real-time chat, deadline reminders, and broader customization options.

V. CONCLUSION AND RECOMMENDATION

A. Conclusion

The development of *MyIntern*, a centralized web-based On-the-Job Training (OJT) management system, successfully addressed the key challenges in the manual OJT workflow at Caraga State University's College of Computing and Information Sciences (CCIS). Through its modular, role-specific architecture, the platform streamlined core processes including student registration, resume creation, job matching, requirement submission, and internship progress tracking.

The integration of structured data flow, resume-based job prioritization, and secure document management significantly reduced administrative overhead and enhanced coordination among students, coordinators, and industry partners. Usability testing demonstrated that *MyIntern* achieved consistently favorable scores across all user roles indicating that it is an effective, intuitive, and efficient system for academic internship coordination.

While the system has proven highly functional, user feedback also suggested several areas for enhancement, reinforcing the platform's potential as a scalable and adaptive solution not only for CCIS but also for other academic departments and institutions.

B. Recommendation

In light of the results, the following recommendations are proposed to maximize the platform's impact:

1. **Institutional Adoption:** The system is recommended for official deployment within CCIS to replace the manual OJT process and standardize internship handling.
2. **User Training and Onboarding:** Conduct orientation sessions and video tutorials to ensure students, coordinators, and employers fully understand system functionalities.
3. **Feature Enhancements:**
 - Integrate **automated reminders** for deadlines and document submissions.
 - Add **real-time messaging/chat** between coordinators and students.
 - Implement **advanced filters** for job searches and applicant screening.
 - Develop **batch processing tools** for requirement validation.
4. **Mobile Optimization:** Develop a mobile-responsive version or standalone app to ensure accessibility for users on smartphones and tablets.
5. **Continuous Evaluation:** Establish a feedback loop post-deployment to monitor user satisfaction, fix bugs, and identify new features based on evolving institutional needs.

Through these enhancements, MyIntern can evolve into a comprehensive, long-term digital solution that significantly improves academic internship administration across the university.

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